SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 8.01

C SAAT Limits, Derivatives, Definite Integrals & Area

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

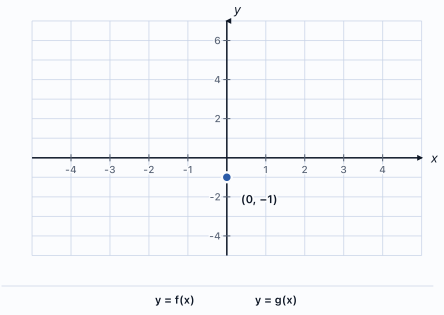
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 B	Q4 C	Q5 D
Q6 A	Q7 B	Q8 C	Q9 B	Q10 C
Q11 B	Q12 D	Q13 C	Q14 C	Q15 D
Q16 A	Q17 D	Q18 A	Q19 D	Q20 D
Q21 D	Q22 A	Q23 B	Q24 C	Q25 B

SAAT Limits, Derivatives, Definite Integrals & Area Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 20x + 118$ in the domain of SAAT Differential & Integral Calculus. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(12)$.



A 50

B 22

C 38

D 40 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Complete the square: $x^2 - 20x + 118 = (x-10)^2 - 100 + 118 = (x-10)^2 + 18$, giving minimum value $k = 18$. **[M1]**
2. Evaluate at $x = 12$: $f_1(12) = (2)^2 + 18 = 22$. **[M1]**
3. Sum the two quantities: $k + f_1(12) = 18 + 22 = 40$. **[A1]**

- **Final Verified Answer:** 40.

2 MCQ

A mathematical model in SAAT Differential & Integral Calculus is defined by the cubic function $y_2(x) = 3x^3 - 10x^2 + 15x - 15$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

- **A** 36 ✓
- **B** 96
- **C** 21
- **D** 22

WORKED SOLUTION & VERIFICATION

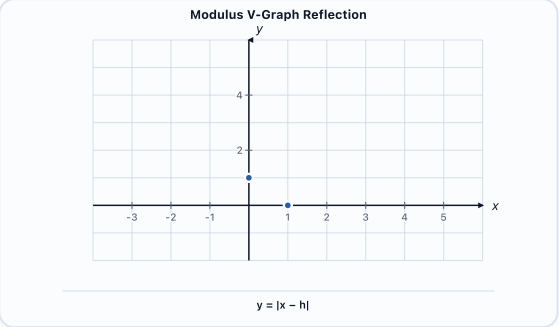
Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 20x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 20(3) + 15 = 36$. **[M1][A1]**

- **Final Verified Answer:** 36.

3 MCQ

In SAAT Differential & Integral Calculus, two displacement vectors are given by $\mathbf{u}_3 = \langle 10, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



- **A** 34
- **B** 198 ✓
- **C** 164
- **D** 310

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (10)(9) + (-8)(7) = 90 + (-56) = 34$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 10^2 + (-8)^2 = 164$. **[M1]**
3. Combine: $34 + 164 = 198$. **[A1]**

- **Final Verified Answer:** 198.

4 MCQ

The linear growth profile shown in the diagram for SAAT Differential & Integral Calculus follows an arithmetic progression with first term $a_1 = 11$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 95

B 885

C 795 ✓

D 1425

WORKED SOLUTION & VERIFICATION

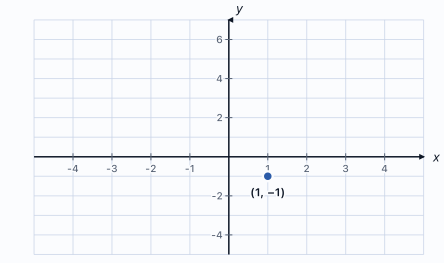
Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 11, d = 6$: $S_{15} = \frac{15}{2}[22 + 14 \times 6] = \frac{15}{2} \times 106 = 795$. **[M1][A1]**

- **Final Verified Answer:** 795.

5 MCQ

In SAAT Differential & Integral Calculus, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 40x)\,\mathrm{d}x$.



A 152

B 168

C 65

D 128 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Integrate term by term: $\int (18x^2 + 40x)\,\mathrm{d}x = [6x^3 + 20x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 20(2)^2) - 0 = 48 + 80 = 128$. **[M1][A1]**

- **Final Verified Answer:** 128.

6 MCQ

In SAAT Differential & Integral Calculus, consider the quadratic function $f_6(x) = x^2 - 22x + 211$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(18)$.

- **A** 229 ✓
- **B** 240
- **C** 139
- **D** 222

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Complete the square: $x^2 - 22x + 211 = (x-11)^2 - 121 + 211 = (x-11)^2 + 90$, giving minimum value $k = 90$. **[M1]**
2. Evaluate at $x = 18$: $f_6(18) = (7)^2 + 90 = 139$. **[M1]**
3. Sum the two quantities: $k + f_6(18) = 90 + 139 = 229$. **[A1]**

- **Final Verified Answer:** 229.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 11x^2 + 33x - 22$ studied in SAAT Differential & Integral Calculus. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

- **A** 609
- **B** 433 ✓
- **C** 400
- **D** 83

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 22x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 9(64) - 22(8) + 33 = 433$. **[M1][A1]**

- **Final Verified Answer:** 433.

8 MCQ

In SAAT Differential & Integral Calculus, two displacement vectors are given by $\mathbf{u}_8 = \langle 11, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

- **A** 53
- **B** 170
- **C** 223 ✓
- **D** 293

WORKED SOLUTION & VERIFICATION

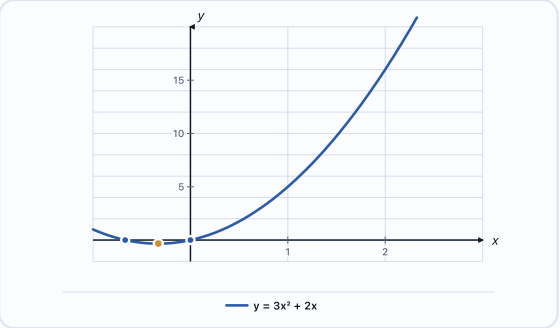
Mark Scheme & Examiner Solution (SAAT Differential & Integral Calculus):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (11)(8) + (-7)(5) = 88 + (-35) = 53$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 11^2 + (-7)^2 = 170$. **[M1]**
3. Combine: $53 + 170 = 223$. **[A1]**

- **Final Verified Answer:** 223.

9 MCQ

Evaluate the definite integral $\int_0^2 (3x^2 + 2x)\,dx$.



A 10

B 12 ✓

C 14

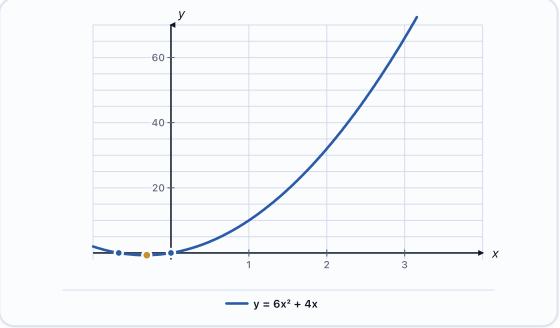
D 16

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 1x^3 + 1x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 12$.

10 MCQ

Evaluate the definite integral $\int_0^3 (6x^2 + 4x)\,dx$.



A 66

B 69

C 72 ✓

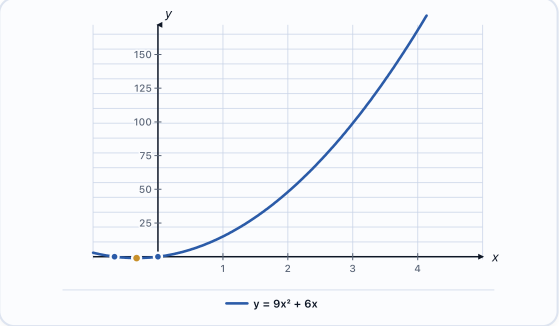
D 76

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 2x^3 + 2x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 72$.

11 MCQ

Evaluate the definite integral $\int_0^4 (9x^2 + 6x)\,dx$.



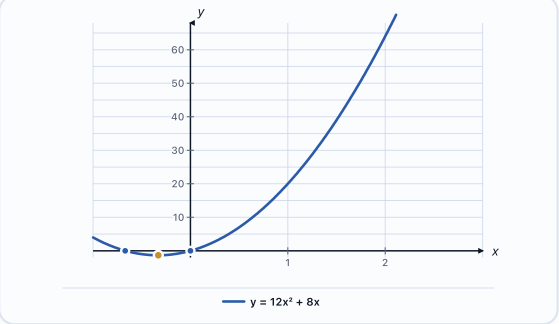
A 168	B 240 ✓
C 246	D 236

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 3x^3 + 3x^2$. By the Fundamental Theorem of Calculus, $F(4) - F(0) = 240$.

12 MCQ

Evaluate the definite integral $\int_0^2 (12x^2 + 8x)\,dx$.



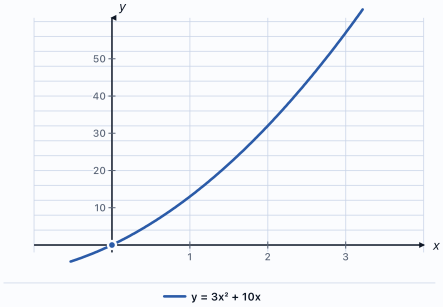
A 56	B 64
C 46	D 48 ✓

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 4x^3 + 4x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 48$.

13 MCQ

Evaluate the definite integral $\int_0^3 (3x^2 + 10x)\,dx$.



A 82

B 69

C 72 ✓

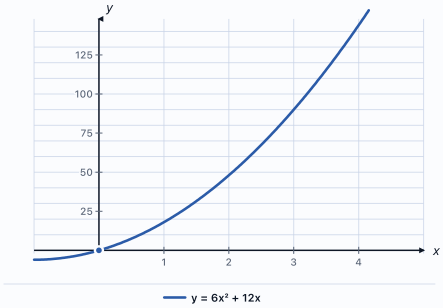
D 57

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 1x^3 + 5x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 72$.

14 MCQ

Evaluate the definite integral $\int_0^4 (6x^2 + 12x)\,dx$.



A 220

B 144

C 224 ✓

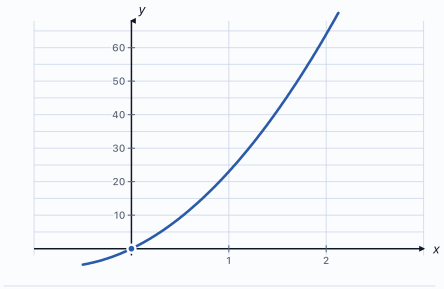
D 236

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 2x^3 + 6x^2$. By the Fundamental Theorem of Calculus, $F(4) - F(0) = 224$.

15 MCQ

Evaluate the definite integral $\int_0^2 (9x^2 + 14x)\,dx$.



A 64

B 50

C 66

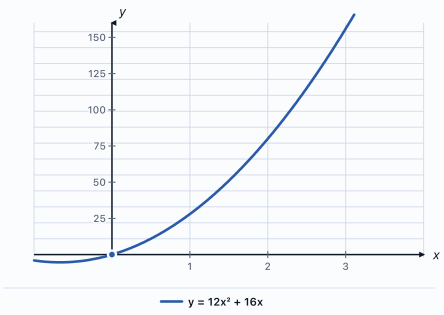
D 52 ✓

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 3x^3 + 7x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 52$.

16 MCQ

Evaluate the definite integral $\int_0^3 (12x^2 + 16x)\,dx$.



A 180 ✓

B 196

C 177

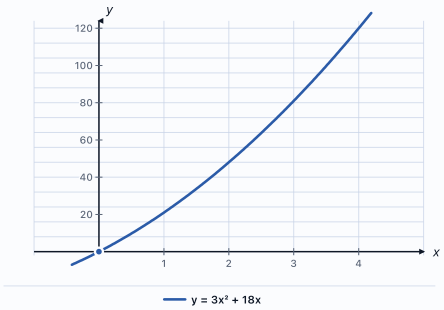
D 156

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 4x^3 + 8x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 180$.

17 MCQ

Evaluate the definite integral $\int_0^4 (3x^2 + 18x)\,dx$.



A 120

B 226

C 204

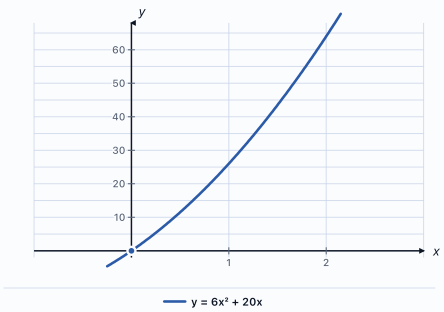
D 208 ✓

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 1x^3 + 9x^2$. By the Fundamental Theorem of Calculus, $F(4) - F(0) = 208$.

18 MCQ

Evaluate the definite integral $\int_0^2 (6x^2 + 20x)\,dx$.



A 56 ✓

B 64

C 76

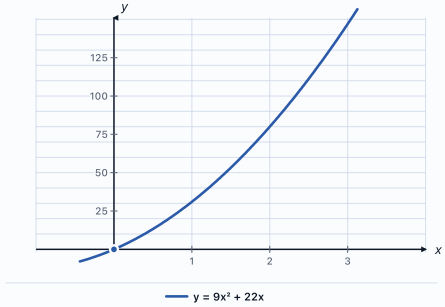
D 54

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 2x^3 + 10x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 56$.

19 MCQ

Evaluate the definite integral $\int_0^3 (9x^2 + 22x)\,dx$.



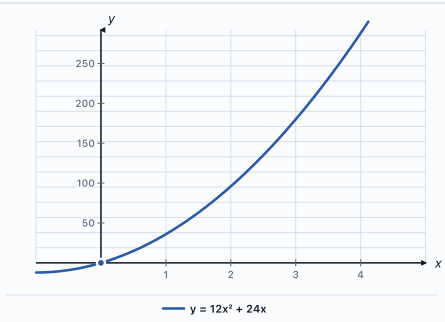
A	177	B	202
C	147	D ✓	180

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 3x^3 + 11x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 180$.

20 MCQ

Evaluate the definite integral $\int_0^4 (12x^2 + 24x)\,dx$.



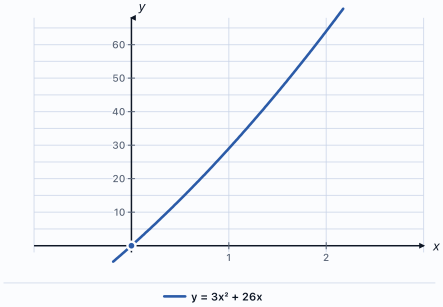
A	444	B	472
C	288	D ✓	448

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 4x^3 + 12x^2$. By the Fundamental Theorem of Calculus, $F(4) - F(0) = 448$.

21 MCQ

Evaluate the definite integral $\int_0^2 (3x^2 + 26x)\,dx$.



A 86

B 64

C 58

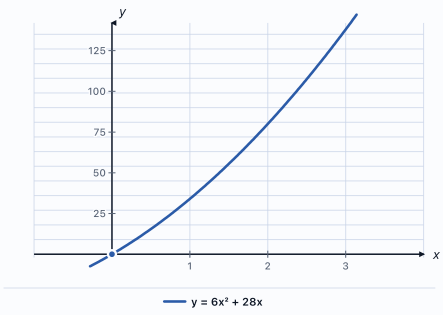
D 60 ✓

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 1x^3 + 13x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 60$.

22 MCQ

Evaluate the definite integral $\int_0^3 (6x^2 + 28x)\,dx$.



A 180 ✓

B 177

C 138

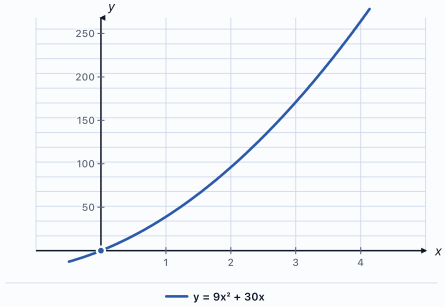
D 208

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 2x^3 + 14x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 180$.

23 MCQ

Evaluate the definite integral $\int_0^4 (9x^2 + 30x)\,dx$.



A 264

B 432 ✓

C 462

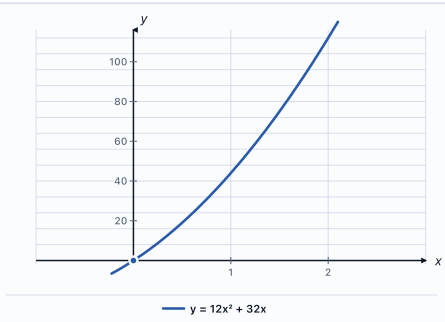
D 428

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 3x^3 + 15x^2$. By the Fundamental Theorem of Calculus, $F(4) - F(0) = 432$.

24 MCQ

Evaluate the definite integral $\int_0^2 (12x^2 + 32x)\,dx$.



A 94

B 128

C 96 ✓

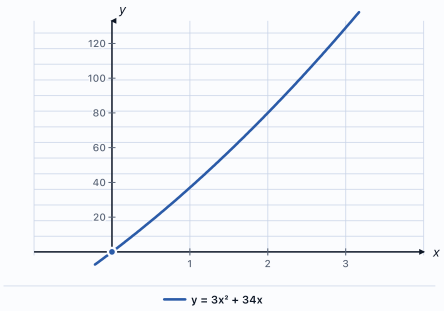
D 112

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 4x^3 + 16x^2$. By the Fundamental Theorem of Calculus, $F(2) - F(0) = 96$.

25 MCQ

Evaluate the definite integral $\int_0^3 (3x^2 + 34x)\,dx$.



A 129

B 180 ✓

C 214

D 177

WORKED SOLUTION & VERIFICATION

An antiderivative is $F(x) = 1x^3 + 17x^2$. By the Fundamental Theorem of Calculus, $F(3) - F(0) = 180$.